OpenStack Private Cloud:

Secure Tenant Deployment & Microsegmentation

Mohsin Rasul

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Abstract

This report details the architecture, deployment, and validation of a highly secure, isolated tenant environment within an enterprise-grade OpenStack private cloud. Driven by the need for strict data sovereignty and minimized attack surfaces, the deployment utilized DevStack on a localized VMware ESXi hypervisor to construct a software-defined network partitioned from the public internet. Compute resources were explicitly constrained using custom hardware flavors to prevent noisy-neighbor performance degradation, while external reachability was decoupled through 1-to-1 Network Address Translation (NAT) via floating IPs. Furthermore, microsegmentation was enforced at the virtual network interface layer using a zero-trust, default-deny security group policy, strictly permitting only necessary administrative and HTTP traffic. The resulting infrastructure provides a robust, reproducible foundation for hosting cloud-native workloads and conducting advanced cybersecurity vulnerability research.

1 Introduction

The record of this experiment begins with the provisioning of an isolated cloud tenant environment on a localized VMware ESXi-hosted hypervisor. The goal of the experiment is to build a functional, access-controlled cloud infrastructure capable of securely hosting virtual machines. This includes expanding physical host storage, configuring stable virtual networking, deploying a zero-trust firewall policy, and verifying external stateful packet filtering.

2 Procedures and Observations

The host environment was prepared by setting Python 3.10 as the default interpreter, clearing legacy DevStack caches, and upgrading system packages. The root logical volume was extended to 100% of the available free space (`/dev/ubuntu-vg/ubuntu-lv`) and resized to accommodate the heavy infrastructure footprint. The DevStack repository (stable/2023.2) was cloned and executed. Due to initial networking hangs, the Open vSwitch and OVN services were manually restarted, and the stack was rebuilt using the `unstack.sh` and `stack.sh` scripts to achieve a healthy operational state.

An Ubuntu 24.04 ISO was transferred to the host via SCP and imported into the Glance image service. A tenant network (`tenant-net`) and a corresponding subnet (`192.168.100.0/24`) were created. A virtual router (`tenant-router`) was deployed, connected to the external public network, and attached to the tenant subnet. A security group (`tenant-secgroup`) was created with explicit ingress rules to allow TCP port 22 (SSH) and TCP port 80 (HTTP). A custom compute flavor (`project-flavor`) was defined with 1024 MB of RAM, 5 GB of disk space, and 1 vCPU. Two virtual machines (`instance-1` and `instance-2`) were instantiated using this flavor and the Ubuntu image. Finally, floating IPs were allocated from the public pool and associated with both instances.

3 Summary of Results

The tenant environment successfully hosted two active instances routed to the external network. Security rules actively filtered traffic, which was verified by performing network probes against the assigned floating IP (`172.24.4.35`). Specifically, `curl` and `nmap -Pn -p 22,80` commands confirmed that ports 22 and 80 were open and responsive, while all other ports were implicitly denied.

4 Calculations and Graphs

The tenant subnet was provisioned with a `/24` Classless Inter-Domain Routing (CIDR) block. This calculates to a total of 256 IP addresses ($2^{(32-24)}$). Reserving the network address, broadcast address, and default gateway leaves 253 usable IP addresses for future compute instance expansion.

5 Analysis of Data and Errors

The reproducibility of the cloud deployment is highly reliable when executing the CLI commands sequentially.

- **Systematic Error:** A major source of systematic error during the initial deployment was insufficient disk space, which caused provisioning failures until the logical volume was manually expanded. Operating on a localized VMware ESXi cluster also introduced minor nested virtualization latency during the parallel instance deployment.
- **Random Error:** DevStack's non-persistent nature introduced random errors related to the Open vSwitch state, necessitating manual service restarts to stabilize the network bridges.

6 Discussion, Interpretation

The outcome of the experiment quantitatively validates the enforcement of predefined compute limits and qualitatively demonstrates functional network isolation.

- **Summary of key results:** The tenant environment was successfully built, segmented from the public network, and secured behind a custom firewall policy, culminating in verified external access.
- **Significance of findings:** Implementing a zero-trust model at the virtual Network Interface Card (vNIC) layer is critical. This microsegmentation prevents lateral movement, ensuring that a compromise of one instance does not expose adjacent workloads.
- **Unusual difficulties:** Storage constraints and Open vSwitch service hangs temporarily halted deployment, highlighting the fragility of single-node sandbox environments.
- **Suggestions for improvement:** To enhance reliability, future deployments should transition to Infrastructure as Code (Terraform) for declarative resource management and utilize Kolla-Ansible for highly available, persistent containerized services.

7 Conclusion

The secure OpenStack tenant environment was successfully provisioned and verified. The deployment enforced strict network isolation and validated external access controls, serving as a functional blueprint for production-scale infrastructure.

References

1. OpenStack Foundation. *DevStack Documentation*. `https://docs.openstack.org/devstack/latest/`

2. OpenStack Foundation. *OpenStack Networking Guide*. `https://docs.openstack.org/neutron/latest/`

Appendices

Appendix 1: Horizon Dashboard Verification

The following telemetry extracted from the OpenStack Horizon dashboard confirms the active state of the instances and their exact compliance with the defined resource quotas.

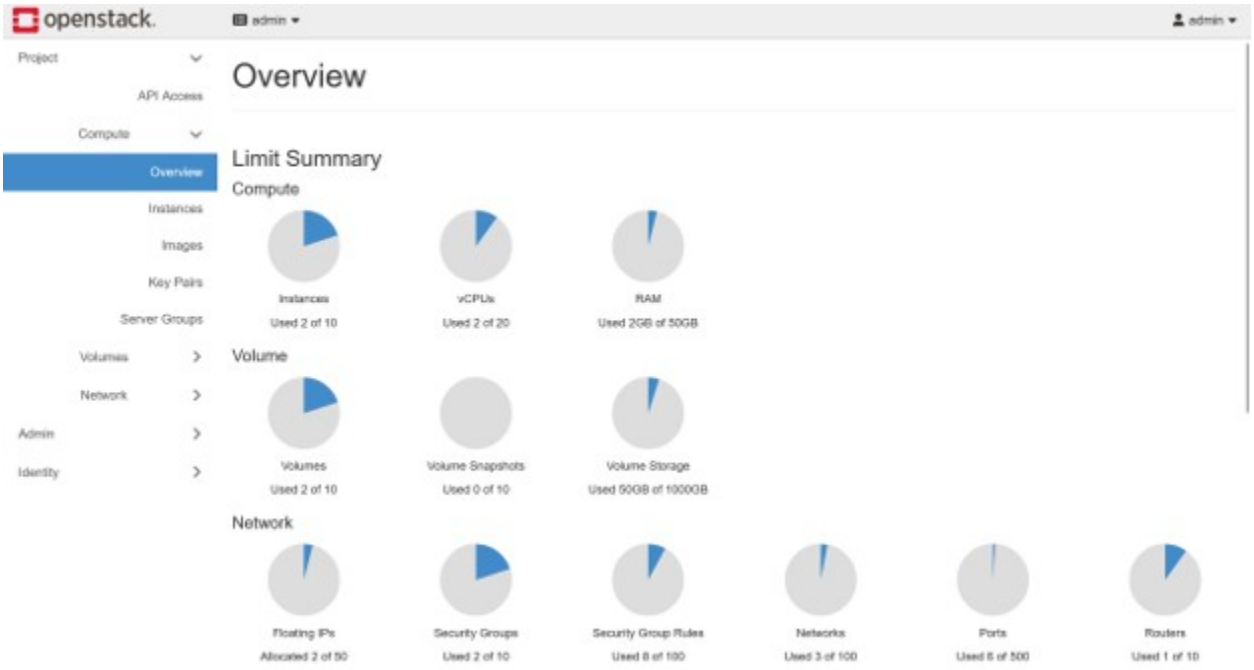


Figure 1: Horizon Limit Summary confirming 2 allocated instances consuming exactly 2 vCPUs and 2GB RAM, matching the custom `project-flavor` constraints.

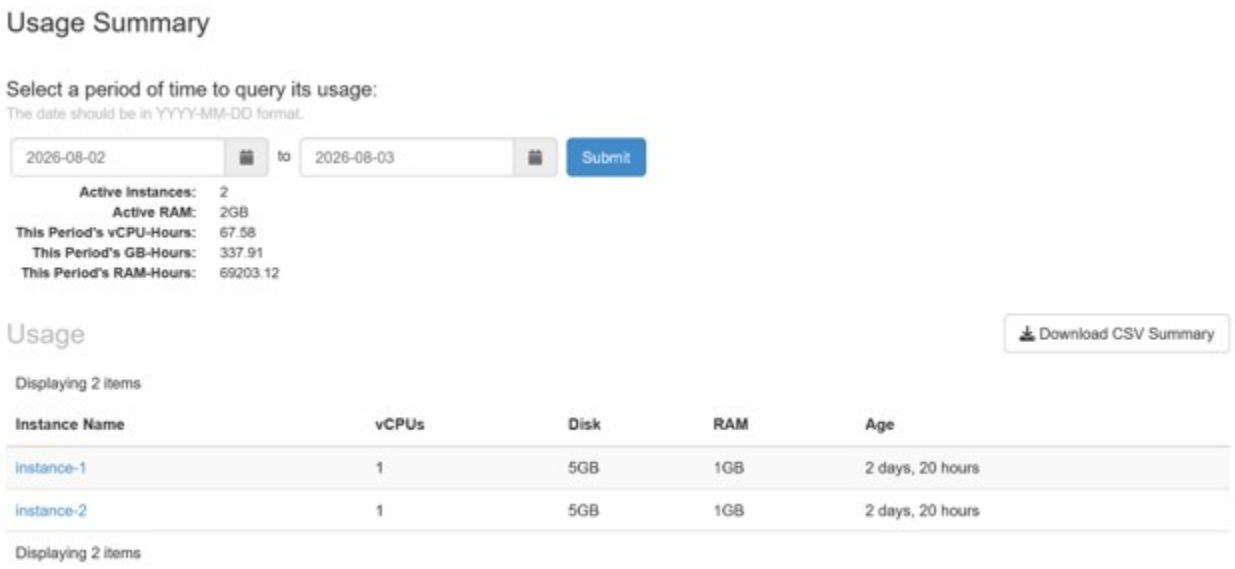


Figure 2: Horizon Usage Summary verifying instance uptime and exact hardware specifications (1 vCPU, 5GB Disk, 1GB RAM per node).

Appendix 2: Complete Bash Command History

```
 1  cd
 2  cd devstack
 3  sudo update-alternatives --set python3 /usr/bin/python3.10
 4  sudo apt-get update
 5  sudo rm -rf /opt/stack/requirements
 6  sudo rm -rf /opt/stack/data/venv
 7  sudo rm -rf ~/.cache/pip
 8  cd ~
 9  rm -rf devstack
10  git clone -b stable/2023.2 https://opendev.org/openstack/devstack
11  sudo apt update && sudo apt dist-upgrade -y
12  sudo apt autoremove -y
13  sudo reboot
14  cd
15  cd devstack
16  git clone https://opendev.org/openstack/devstack
17  cd devstack
18  nano local.conf
19  ./stack.sh
20  sudo shutdown now
21  cd
22  ./unstack.sh
23  cd devstack
24  ./unstack.sh
25  sudo systemctl restart openvswitch-switch
26  sudo systemctl restart ovn-central
27  ./stack.sh
28  source openrc admin admin
29  openstack image create "Ubuntu-VM"   --file ~/Downloads/ubuntu
-24.04.4-live-server-amd64.iso   --disk-format iso   --container-format
 bare   --min-disk 15   --min-ram 2048   --architecture x86_64   --
public
30  openstack image create "Ubuntu-VM"   --file ~/Downloads/ubuntu
-24.04.4-live-server-amd64.iso   --disk-format iso   --container-format
 bare   --min-disk 15   --min-ram 2048   --property architecture=x86_64
   --public
31  scp "C:\Users\Mohsin Rasul\Downloads\ubuntu-24.04.4-live-server-
amd64.iso" stack@192.168.253.130:~/Downloads/
32  scp "C:\Users\Mohsin Rasul\Downloads\ubuntu-24.04.4-live-server-
amd64.iso" mohsin@192.168.253.130:~/Downloads/
33  exit
34  cd
```

```
35  ls
36  exit
37  ls
38  cd
39  ls
40  /home
41  cd /home
42  ls
43  cd
44  ls -la
45  mkdir -p ~/Downloads
46  sudo ls -lh /opt/stack/data/glance/images/
47  ls
48  cd Downloads
49  ls
50  cd
51  cd /opt/stack/data/glace/images/
52  cd /opt/stack/data/glance/images/
53  ls
54  cd
55  sudo passwd stack
56  exit
57  rm -f ~/Downloads/ubuntu-24.04.4-live-server-amd64.iso
58  rm -f ~/ubuntu-24.04.4-live-server-amd64.iso
59  sudo lvextend -l +100%FREE /dev/ubuntu-vg/ubuntu-lv
60  sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv
61  df -h /
62  cd devstack
63  ls
64  cd
65  ls
66  cd Downloads
67  ls
68  cd ~/devstack
69  source openrc admin admin
70  openstack image create "Ubuntu-VM"   --file ~/Downloads/ubuntu
-24.04.4-live-server-amd64.iso   --disk-format iso   --container-format
 bare   --min-disk 15   --min-ram 2048   --property architecture=x86_64
   --public
71  cd
72  cd devstack
73  source openrc admin admin
```

```
 74  openstack image create "Ubuntu-VM"   --file ~/Downloads/ubuntu
 -24.04.4-live-server-amd64.iso   --disk-format iso   --container-format
  bare   --min-disk 15   --min-ram 2048   --property architecture=x86_64
    --public
 75  openstack network list
 76  openstack image create
 77  exit
 78  cd devstack
 79  ls
 80  ./stack.sh
 81  cd devstack
 82  ls
 83  ./stack.sh
 84  ./unstack.sh
 85  ./stack.sh
 86  cd
 87  ls
 88  mkdir testVM
 89  ls
 90  cd testVM
 91  nano key.pem
 92  ls
 93  ssh -i key.pem ubuntu@192.168.233.121
 94  chmod 400 key.pem
 95  ssh -i key.pem ubuntu@192.168.233.121
 96  ssh -i key.pem ubuntu@172.24.4.127
 97  ssh -i key.pem ubuntu@172.24.4.99
 98  exit
 99  cd testVM
100  ls
101  nano key.pem
102  cd
103  cd testVM
104  ssh -i key.pem ubuntu@172.24.4.127
105  ls
106  nano key.pem
107  cd
108  cd devstack
109  ./unstack.sh
110  sudo shutdown
111  cd devstack
112  source openrc admin admin
113  openstack network create --network tenant-net
```

```
114  openstack network create tenant-net
115  openstack subnet create --network tenant-net --subnet-range
 192.168.100.0/24 --dns-nameserver 8.8.8.8 tenant-subnet
116  openstack router create tenant-router
117  openstack router set tenant-router --external-gateway public
118  openstack router add subnet tenant-router tenant-subnet
119  openstack security group create tenant-secgroup
120  openstack security group rule create --protocol tcp --dst-port 22
 tenant-secgroup
121  openstack security group rule create --protocol tcp --dst-port 80
 tenant-secgroup
122  openstack image list
123  openstack flavor list
124  openstack flavor create --ram 1024 --disk 5 --vcpus 1 project-
 flavor
125  openstack flavor list
126  openstack server list
127  openstack server delete 2b9969c1-1596-4eaa-9808-1e0fb67eb09f
128  openstack server create --flavor project-flavor --image bee5c342-15
 a6-4ade-bca5-f1a7c67fb0d4 --network tenant-net --security-group tenant-
 secgroup instance-1
129  openstack server create --flavor project-flavor --image bee5c342-15
 a6-4ade-bca5-f1a7c67fb0d4 --network tenant-net --security-group tenant-
 secgroup instance-2
130  openstack server list
131  openstack floating ip create public
132  openstack server add floating ip instance-1 172.24.4.35
133  openstack server add floating ip instance-2 172.24.4.174
134  openstack server list
135  curl http://172.24.4.35
136  nmap -Pn -p 22,80 172.24.4.35
```